

ActInf Livestream #007.1: "Variational ecology and the physics of sentient systems" (2019)

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all right hello and welcome everyone to the active inference live stream this is active inference live stream 7.1 and it is october 27th 2020. welcome to team com everyone we are an experiment in online team communication learning and practice related to active inference you can find us on twitter at inferenceactive at our gmail you can find us on key base or youtube as well this is a recorded and an archived live stream so please provide us feedback so that we can improve our work all backgrounds and perspectives are welcome here and as far as video etiquette for live streams mute raise your hand so that we can get to everyone in the queue use respectful speech and try not to use the jitsi chat if possible today in act imp 7.1 we're going to start with a warm-up just briefly introducing ourself and checking in and then we'll work through some of the sections of 7.1 the paper we're going to be discussing is variational ecology and the physics of sentient systems by ramstead et al in 2019 and today we'll talk about the goals abstract the roadmap potentially move through a few key quotations and then preview the figures mainly with a focus on seeing what each figure represents and today in 7.1 and next week in 7.2 where we may or may not be in costume we will discuss this paper so please save and submit your questions whether you're on the panel now or whether you're watching in replay and get in touch if you'd like to participate so for the intros and check-ins it'd be great if everyone could just introduce themselves and their location say hello in a short intro and then pass it to somebody who hasn't spoken so i'm daniel i'm in california and i'll pass it to a first time discussion cameron hi yeah i'm cameron um i'm here in zurich in switzerland i just started a phd in philosophy of mind at the university of zurich first time at this group um shannon hi everyone i'm shannon i'm a phd student at the university of california in merced um i'll pass it to lee hi everyone uh i'm in london a phd student at the university of york um a bit of an active inference beginner and i will pass it to alex hello uh i'm in moscow russia i'm a researcher at systems management school yeah and the opposite session hello um uh my name is sasha i'm in california and i'm a neuroscience graduate student

um and i will pass it to muddy hi i'm muddy um i'm uh the director of ai at uh versus labs in the spatial web foundation uh my link to active inference is uh i i did my phd with karl riston four or five years ago still very much a beginner and i'm currently in uh rainy liverpool and so i will pass it to anybody who would like to volunteer because i can't see the screen i apologize um hello it's stephen here i'm in toronto i'm doing a practice-based phd at canterbury christchurch university in the uk and i'll pass it over to maxwell hey everyone i'm maxwell ramstead i'm a postdoctoral fellow at mcgill university in montreal i'm skypeing to you from the montreal suburbs um and yeah i work on uh multi-scale extensions of active inference i'm the um first author on on this paper that we'll be discussing uh very grateful to have such a great group of people as discussants and i will pass it to i believe lee is the only one who hasn't spoken yet right no no i've been cool yeah i think that gets through it perfect sasha has no yeah she has okay sorry yep great thanks everyone for the intros so in these two warm-up questions any thoughts related or unrelated anything that kind of gets you excited and helps on-ramp to the discussion we'll be having what is a physical or informational ecosystem that you care about or have studied and while people are raising their hand um the first ecosystem that came to mind for me was in southeast arizona where there's a grassland ecosystem where there's a lot of cool ants and also the grazing agricultural use by cows had a big effect on the ecosystem kind of pushing it from a grassland regime into a very um different sort of pebbly regime so when i was thinking about this variational ecology i was kind of thinking about the desert ecosystem and and the southwest and go ahead muddy um so i'm interested in a slightly different non-biological ecosystems uh but still ecosystems where we have agents that make decisions so we build our micro service meshes by having multiple agents that can make decisions and can decide for themselves whether they are appropriate to a given computation or not so we treat them as ecosystems and we are using active inference very much to model the the life cycles of these various micro services cool and if anyone else wants to raise their hand muddy i'm actually just kind of curious let's go to steven first and then i'll follow up if possible yeah i'm interested in kind of intentional spaces but learning spaces or workspaces and one thing that really came up for me was when i was in a place called inguavuma at poncini school back in 2004 when i visited there and to do a project or just at that time just to engage the community i it was at a school then ponchini school in a rural area where they'd only just built the school 10 years before so they didn't actually have schools there before that and um in in and you know around the outside you've got the fence with the barbed wire on the top in the middle of this area with no electricity and it was and i i i sensed at that time there

was this kind of disconnect between the students understanding what this niche was that they were immersed in and an area where um you know while they're learning stuff there's also 35 hiv and big challenges and and this kind of learning environment which was kind of coming from a colonial background so i this this kind of difference between the world which i was seeing in that community in the world in this this school kind of you know as i've seen in certain workplaces you see this kind of disconnect this kind of like learning isn't just the knowledge it's something about how to fit it together and that's when i so later became this idea of a social topography whether somehow what's that kind of landscape of meaning and when is it more than just knowing the facts and i think that's where the sort of comes together cool i like how we're exploring a lot of these types of ecosystems and those are definitely what we want to encompass in the variational ecology discussion sasha um yeah so i'm interested in the micro environment that neurons grow up in and how that affects their activity levels and how they make connections with others so one of the things that i'm studying for that is how different immune molecules and turning on immune genes affects this process and this is something that i want to learn from this paper um how to apply that to my own system but one parallel i'm also thinking about is uh the immune landscape um when people are isolated and aren't exposed to as many people what their kind of immunity is like um especially in a time when we're really thinking about our own health and immunity ooh just from writing down some of these little notes there's ecosystems across scale that's definitely going to come back and then there's this idea of community ecology which in the bio sense is usually about different species but then there's of course as steven was talking about this group and historical dynamics to our community ecology and then ecosystems that are of different technologies so the second warm-up question is what is something that you're curious about or want to explore in today's discussion and while people are raising their hand i think the question that almost motivated or seems to motivate this paper is the one i'm curious about which is about applying the free energy principles formalisms to larger scale systems where potentially the stability of the markov blanket or some of the other dynamics of the system are so rapidly changing that it um really questions how we're going to study it from this framework and i know that's something that we'll return to muddy yeah i i think what's fascinating um from from this work that maxwell has been doing is is helping us actually get really fine-grained detail onto the computational currents of certain problems such as you know when do i stop modeling my subsystem and when do i start representing my subsystem as a parameter so when you know this transition between a parametric approach and a modeling approach to then feed into a

hierarchical model that is uh quite a it's quite an art and it's an ill-posed sort of problem so i think this work is really really fascinating from that perspective from me personally cool maybe maxwell i'd be curious what is that relationship with the parametric and the modeling approaches like how do we go about exploring that or what does that mean from a modeler's perspective i'm not sure i understand your question um like muddy was talking about how this uh question of modeling like when to stop and when to include a parameter at one level versus another what's an introductory thought or what's the rationale or the motivation behind doing that kind of a thing i mean so mel andrews has thought about this a lot more than i have i mean uh i wish they were here um that they're sick today so i mean roughly speaking the generative models that we employ under active inference uh they harness the what mel has been calling the existential variables of the uh so you know the the variables that actually make a difference to uh survival uh so you know you can think of stuff like a core body temperature and so on i mean ultimately there is always going to be an arbitrariness uh involved in the modeling effort because we have to make kind of science science side decisions about what are the relevant things that we're going to model or not but that said there do seem to be some some variables that are more critical to the continued existence of the systems that we're interested in so i don't know if that addresses the point cool i think we'll return to it especially as we think about what are the vital signatures or the vital variables for ecosystems because we want to be thinking about them in similar framework cool here we go so the paper that we're going to be discussing in 7.1 today and in 7.2 last week as well as in 7.0 last week uh yeah people can figure that one out variational ecology in the physics of sentient systems by ramstead constant badcock and firsten and the goal of the paper was stated as reviewing a framework for modeling complex adaptive systems for multi-scale free energy bounding organism niche dynamics thereby integrating the modeling strategies and heuristics of variational neuroethology or vne with a broader perspective on the ecological nestedness of biotic systems we extend variational neuroethology beyond the action perception loops of individual organisms i.e active by appealing to the variational approach to niche construction to explain the dynamics of coupled systems constituted by organisms and their ecological niche and so i quoted that directly from the paper because i think it's a really direct representation and foreshadows the fusion of the vne and the va nc which we can talk about but to kind of rephrase that in non-fep language it's a question about how can we make a multi-level or multi-scale ecology framework that builds upon eco evo devo that's ecology evolution and development but also adding in insights from physics mathematics and complexity or complex

adaptive systems so it certainly is an ambitious scope and it's kind of linking together this organism and downward story of the variational neuroethology with the niche construction aspect so kind of we have the organism looking downwards that's the neuromolecular components of behavior and then we have the organism looking upwards and outwards with the ecology yeah if i could comment on that briefly yeah so the basically the if you're looking at like the history of how this came about um so i was working i mean between 2000 i guess 16 and 18 i was mostly focused on developing uh you know this active inference story in large part with carl fristen and that resulted in a framework that we called a variational ethology or variational neuroethology in our 2018 physics of life reviews paper um answering short ingress question free energy formulation and so the the variational neuro mythology is essentially uh a theory about the kind of if you if you want to think about it like um visually a kind of vertical stack of systems we've discussed this on this podcast before but like this idea that systems are made of nested systems of systems so that essentially if you look at the ontology of a system what you have is a stack of segregated systems of different spatial spatial and temporal scales that affect effectively compose each other so cells composing tissues composing organs composing organisms composing social groups and composing species and so on and so following the publication of that paper i started working closely with axel constant who uh was riffing off work that yella brinaberg had been doing in amsterdam uh with his group over there and uh axel and i developed um a basically a niche construction account of active inference what we highlighted was the fact that active inference is a symmetric formulation in the sense that it tells a story about how internal and external states attune to each other internal states of tuning to external ones through things like perception and and learning and phenotypic accommodation and development and external states tuning to internal states through action notably making the world more like uh the way that we expect it to be so that gave us a kind of horizontal dimension where basically at every kind of level of this uh vertical stack which you have is effectively a relationship between uh an ecological niche and the denizens or inhabitants of that ecological nation so this paper uh aimed to combine both frameworks uh so the niche construction stuff and the nested systems of system stuff into a kind of principled uh approach to biological systems uh and so you know moving from an ethology and a theory of niche construction so ethology being the study of animal behavior essentially so combining these two things into a bona fide theory of ecology uh using the variational free energy principle i'm sure you're going to mention this after but the last ingredient that we added to this mix is the skilled intentionality framework from the

amsterdam group eric groupveld julian kieverstein uh yellow greenenberg what they add to the mix with the skilled intentionality stuff is connecting this predictive processing framework to the ecological psychology notion of affordances um yes so this paper essentially takes the ecological psychology stuff on affordances the so-called skilled intentionality framework the variational ethology and the variational niche construction stuff and proposes this integrative model uh yeah so that's the kind of history awesome great summary thanks so much maxwell and just to kind of restate that in this nested framework you have the bottom up or emergence features of the system you have the top-down features of the system and then the lateral are kind of like the collective so we're thinking about all these ingredients that maxwell just mentioned especially coming from the skilled intentionality framework and a few other areas bringing them all together so let's check out the abstract this paper addresses the challenges faced by multi-scale formulations of the variational or free energy approach to dynamics that obtain for large scale ensembles such as ecosystems or groups of organisms we review a framework for modeling complex adaptive control systems for multi-scale free energy bounding organism niche dynamics thereby integrating the modeling strategies and heuristics of variational neuroethology which remember was really focused on the organism's feedback with its local affordances with a broader perspective on the ecological nestedness of biotic systems so like a lot of times in this podcast we've talked about the affordance of getting a jacket or of changing the thermostat but that's actually part of the constructed niche of humans it's not just a human in a box it doesn't find themselves without affordance unlike these some of the biomechanical affordances perhaps we extend the multi-scale variational formulation beyond the action perception loops of individual organisms systems constituted by organisms and their ecological niche and that relates to the symmetry that was being discussed a few minutes ago we suggest that the statistical robustness of living systems is inherited in part from their eco-niches as niches help coordinate dynamical patterns across larger spatiotemporal scales we call this approach variational ecology so it's kind of like the affordance of driving in a car is part of the constructed niche of the highway network and then that shapes the movement patterns we argue that when applied to cultural animals such as humans variational ecology enables us to formulate not just the physics of individual minds but also a physics of interacting minds across spatial and of physics of sentient systems that range from cells to society and this should sort of invoke that warm-up discussion about the different kinds of ecosystems especially when talking about humans and our technologically enabled extended niches part of the ecosystem is informational it's symbolic and so we also

want to be thinking about information ecologies in the way that these dynamics apply not just to the ants on the pebbles but also to humans and the paths through the parks that are discussed in the and also ways that long-range technological communication might come into play so just to briefly run through the road map this paper is zero indexed which i found very interesting and it starts with a of the variational or free energy formulation and just briefly reviews active inference and generative models it's really great that every paper sort of starts with this uh section on just what it is that we're talking about with active and fvp they're so annoying to write though honestly you know you don't want to you don't want to plagiarize you don't want to just copy the paragraphs you've written uh it's like how many ways can you explain the same thing you know and it's like a if there really is a smooth energy landscape underlying these explanations we're treading many many paths and exploring many on-ramps turning these ideas around through discussion and through more papers so yeah there's a repetition element and there's also potentially a way that we make bigger and more accessible explanations by rephrasing it again and again for different journals or different specialties so then the second section is on variational neuro anthology and that brings into question uh multi-scale levels of analysis as well as ensembles of markov blankets as well as a few figures that we'll scan through the third section introduces the sort of organism and upwards perspective that is given by the niche construction the variational approach to niche construction and relates that to ontologies of affordances some of which we've talked about in previous discussions as well another figure on the bayesian mechanics and active inference and then as usual with these papers there's sort of the foundation then idea a and idea b and then i a and b are linked together and so we get to section four variational ecology the physics of shared mites and this also relates to discussions we've had about what it is that's modeling is the organism a model of its environment does have a model and agreed that mal's thoughts on this are super helpful and then there's some concluding remarks and just to really quickly visually outline this theory by addition approach that's being enacted here on the very bottom we have the fep which is that paradigmatic ground rock that a lot of these other theories are going to be built on top of then from left to right we have active inference tin bergens four questions which are explored a little bit in number 7.0 as well as some previous discussions we've had est is evolutionary systems theory which is basically evolutionary biology with a systems perspective and eco evo devo which is also a very interesting and rich area that emphasizes that the way that evolution occurs is not just ecologically embedded and enacted but it occurs through changes in development so you don't just get a longer femur you get the

developmental proclivity to have your growth plate be open for a longer or shorter time resulting in a phenotype that is changing but it's changing through development active inference and tin broken for questions and evolutionary systems theory were basically summated this is not a perfect summation it's just sort of building it up a little bit to vne which was a very organism and biobehavioral centric approach to explaining action perception loops of organisms on the other hand also drawings from actinf was the variational approach to niche construction which drew a little bit more extensively from the eco evo devo motivating questions and literature and then to bring in synergetics which is this relationship between the higher order uh parameters the bigger slower things and then the lower level parameters the faster and smaller things of hakan 1983 on the left side uh and then the collective behavioral approach these lateral relationships between nested systems it all kind of comes together to get to variational ecology which hopefully builds upon the strengths of all of these different antecedent theories and gives us something that is going to allow us to dip back into for example behavioral explanations of organisms or dip into the niche construction approach or the collective behavior across different levels of analysis as well as providing a helpful path back to understanding how previous formulations for example a modern synthesis type evolutionary biological explanation or a non-fep eco evo devo explanation it gives us an interpretation path that we can walk to understand a lot of the previous literature so let's just pause here does anyone have any thoughts on this theory summation or is there any other areas that we think might be relevant to draw in one that i was wondering about is this introduction of the minds and the sentient systems and so there was a sentence in the paper about the sentient systems being sensory systems though often when people talk about the search for sentient life and things like that they're talking about the experiential dynamics as well and so the idea of getting the interacting minds from the interacting evolutionary systems i just thought that was kind of a curious thing maybe foreshadowing in integration with psychology or with phenomenology well so the the sentience aspect is really i think just to highlight what the the free energy principle stuff does i mean at the end of the day uh the free energy principle shares with perceptual control theory the idea that action is about uh the control of perceptual input namely keeping it within certain bounds um so that might be relevant to uh to consider i mean yeah so from that point of view sentience is uh the capacity to be affected uh by the world in a way that allows us to control how the way the world affects us yep like something hitting a rock still could be felt or is is still as influential on the state of the rock but it doesn't necessarily have the ability to then act especially in a proactive way to get around

that and also just one other note is actually the schrodinger's question from 1944 and then the answering schrodinger's question paper from 2018 i think are also great places for people who are kind of coming at this from a first principles of biology perspective because schrodinger's question was about how organisms are organized and very interestingly and nine years before the discovery of the structure of dna and many years before a lot of the other approaches that would come on the scene with a molecular revolution in the second half of the 1900s schrodinger said there has to be something like a quasi-periodic crystal and organisms have to organize at least locally they're not going to violate the laws of physics this isn't vitalism but at least locally you have to be able to take you know the bag of sugar and then organize it into biomolecules that constitute an organism that's a increase in order how does that happen is that the defining principle of life and in some ways as we look on earth and elsewhere for life these two ideas of schrodinger really uh remain as prescient as ever so i find that just really interesting stephen i know you're probably going to mention this but i suppose the idea of ergodicity sort of comes in through the active inference peace more than it normally thought of in the other areas but that maybe synergetics isn't another way that these kind of ergodic um ways to extract information at present and i suppose the question is is that non-ergodic processes the ways to tap into things which are non-ergodic and sort of causal is that something which is when you then step outside of normal active inference and you have this extra structure on top which is our cognition which allows us to sort of process things that are kind of non-ergodic so i think that might be interesting like in terms of how we construct things as humans as opposed to that kind of more organic niche construction but that not that that's really touched on here but i suppose it's just just the idea of how patterns are extracted and what levels of meaning or what processes of like attuning or constructing or relating what what's that kind of dynamic at play thanks for that comment i think the word ergotic is really what a lot of that idea hinges on so if anyone wants to have a take at what the word ergotic means i tried to go through it a little bit in 7.0 but um if anyone wants to explain what an ergotic system is and stephen what your question gets at is really whether the models of ergotic systems of systems where the time average is equivalent to the ensemble or the spatial averaging um though there's also other ways to phrase that or perspectives on that in systems with extreme historicity like niche construction if you build a road in one place the whole society kind of leverages around that or if there's an introduction of a new species into an area that can irreversibly and extremely historically change the trajectory that so how are we to model these systems using ergotic frameworks stephen and then anyone else who wants to step

in on the ergodic question because it's kind of a key term yeah i was i think this is a really interesting i thought of a relevant because the thing with ergodicity is i i sort of was looking at the video for this session that you made and sort of comparing it to what i've been thinking is it could be that there's lots of small patches or periods or blocks where there's kind of enough ergodicity due to mixing to extract information or to find patterns so it could be that there's not this idea that the system is ergodic but there might be lots of times where there's moments of mixing that ergodicity is present and you can extract or make inferences from that so i think that's quite um that changes things a little bit in terms of how you know makes it more plausible maybe for how ergodicity can creep in um and it also i think then starts to give you a rationale for how you can engage with the whole entropy question as well and because normally entropy is brought into this equation talking about multiple levels and things dissipating and suddenly it's like well um if you've got periods of ergodicity then you can you you've got a way to extract or get a grip on the noise that's going on out there over time um and and i think that that that's important i think that's also important in terms of trying to explain it to other people because most people when they think about the world and systems i'm actually there's a whole debate on the vibrant group about this is that they kind of they just take a snapshot of perspective on a system and they say this is what the system does but that's not the same as something which is happening over time and there's a pattern over time it's like it's a different way of thinking about the world and even think about systems in general so that that that piece is really important about how does it happen over time and is it lots of bits of ergodicity that add together or is it like we have to have ergodicity over 100 years over 10 years over five years nice let's go shannon and then muddy and then maxwell if you want i i would just like to jump in quickly just to clarify a few things about ergodicity if that was okay uh let's go yeah you might actually answer my questions before i ask them perfect shannon muddy go ahead okay so ergodicity from a mathematical point of view you can understand it heuristically as you know so a system is ergodic if the average of our measures of the converges to a measure of the average of our the value of whatever we're measuring so another way of thinking about it is that like wherever your system starts in its state space with an ergodic system it will end up in in a regime of characteristic states like so so no matter where you start you end up uh in in these uh in these states so i mean you know you can think of uh you know my the my my body temperature for example would be describable in this way you know it varies a bit but it ends up continuously converging to 36 and a half degrees celsius so there's a lot of confusion in the literature and in discussion about this ergodicity thing um one

thing to point out is that uh so to the extent that it is implied in the framework uh the the notion that we're considering is only local ergodicity so we're not saying that uh you know because the maximal version of like an ergotic system like a system that's globally ergodic is a system that doesn't really have a history in the sense that it doesn't really matter where you start you always end up in the same place um the notion of ergodicity that we're employing in the free energy framework to the extent that it's employed at all is the notion of local ergodicity which means that uh on some appropriate temporal and spatial scale uh it's useful to think about the system as being ergotic as in that it will regularly converge to the same uh set of states or values of those states rather um so i guess there's a lot to say here on the one hand um you know this local ergodicity thing uh is important so when we say that the earth is uh is round you know uh we're we're not saying that it's not flat enough locally for us to build buildings on it uh so like locally the the earth has it has fl is flat enough has a negligible enough curvature that we can build for example a building on it and this is the same kind of idea so the the ergodicity at play is that you know on some appreciable time scale systems converge to the same set of values effectively and uh yeah so i guess that's uh one point another related point is ergodicity actually isn't um they well it's it's not really baked into the free energy principle all you really need uh to get the free energy principle running is a markov blanket and a non-equilibrium steady state or generative model which we've been discussing over the last few weeks so generally notions like ergodicity and weakly mixing also appears in some of these discussions it only applies to certain kinds of systems with multiple multiple particles under special assumptions namely the assumption that all the particles are exchangeable um so if we're dealing with a single particle this kind of probability distribution in general doesn't exist the only probability distribution uh to which we have access is when we sample the state of a single particle over time um so yeah the the real assumptions on which the free energy stuff is built is this non-equilibrium steady-state assumption um i mean and it's related to the ergodicity thing in this in a certain sense um so non-equilibrium steady state we're we're saying that you know the system has a set of characteristic states to which it it tends you know over time uh and so if a system has uh a non-equilibrium a steady state then you know under the conditions that we just described or there are particles that are exchangeable then uh you can treat the system as ergotic but it's it's not it's not as crucial to the framework as some would suggest so like i i've heard in the you know in the bushes that uh some people are mounting a critique of the free energy principle based on the idea that uh living systems aren't actually i think that's a bit of a waste of time because we're not we're not

claiming biological systems are globally ergodic just that it's useful to treat them as their gothic uh at certain scales and you know the way that we get out of this kind of weird place formally speaking is by pointing to this nested systems of systems thing so the idea is that from the perspective of any layer of the system the layer above it might as well be stationary so from the point of view of my body of my cells for example my body is essentially stationary you know so their entire life cycle you know from birth to uh cell division uh will occur without any major structural change happening to the non-equilibrium steady state of my whole body so you know you you get around the implausibility of the strong interpretation of these assumptions by pointing out that well what's actually at play is a nested series of systems and you know you only need things to look ergodic locally relative to other layers of the i think that's really helpful for what my question would have been which would be like how would if you're talking about the behavior of a crowd and if there's a certain markov blanket that can extend over this this group of individuals um the thing about a crowd is it's not a cell that's not bounded by being stuck inside of a body they could leave like they could leave the stadium or leave the protest but if we're talking about ergodic or like interacting just enough at just the right spatial temporal scale then we can talk about some markup blanket over some whole crowd or subset of a crowd at some event at some time exactly yeah that's precisely what's going on here so it's all about defining the relevant spatial and temporal scales i mean if your relevant spatial and temporal scale is like you know the whole city over several weeks then obviously the the protesters that gather you know and spontaneously form a social group wouldn't count as you know even having a non-equilibrium steady state i mean it just kind of forms and dissipates but if you're looking over the scale of you know minutes to maybe hours then it does make sense to say well locally you know there's something robust here it has a statistical independence from its environment it seems to be you know attracted to the same set of states and then it dissipates so so it's the same story over and over again just a different kind of spatial and temporal scale so you're totally right i think shannon that's right right on point cool let's do muddy and then stephen uh yeah i just wanted to speak to some of the like just to give a little bit of intuition so i'm kind of not super in active research right now so these terms i i like sharing kind of the way i think about them for people who might be listening who might not be familiar with the like ergodicity in general so just generally i i and please correct me if i'm wrong about this because i'm looking to have my understanding probe but i view ergodicity or a system that's got it um my intuition is that if i take a sample of a system that system is ergodic if that sample is representative of the greater

dynamics of that system so that's what maxwell was saying about tending to the mean so it doesn't really matter if i take my data from here here here or here as long as the sample i'm taking is representative um that there's another camera worms it should tend to the same mean or you know basically represent the same dynamics of the underlying model and underneath but the way in which you define ergodicity is actually defined by your your measurement your sample space it's necessarily defined by your sample space so you have all of these emergence properties which are hard to pin down so you have like dynamic ergodicity which which are you know partials with respect to time which tend towards some stable value for the local and then the actual mean versus static ergodicity which is if i take a snapshot of my system at any one given time so like a cloud of gas it kind of looks the same but without any knowledge of my attractor systems i can't say that my system is ergodic dynamically over time um and then you can think about continual ergodicity at different levels so um i think the example maxwell gave with flat earth is a really really beautiful one because if you take a local representative sample say 100 meters by 100 meters you you could define every single sample by a single function uh a stable function which either has either a stability maximum point or is linear in its nature so at that level i have a continuous function which defines ergodicity across all scales but it's part it's nested within a larger system which might have chaotic dynamics actually and so it's it's actually ergodicity i think the ambiguity comes in when you're actually trying to define the problem space itself as a concept it's just basically saying this sample represents the model i'm trying this sample of a larger system represents the largest system at whatever mathematical perspective i happen to be looking at it from whether it's with respect to time or some other variable so i just wanted to add that in i mean just to play devil's advocate thank you buddy for that i agree with everything you said uh just to make the uh you know put some meat on the bones of criticism and the interest of fairness one of the reasons that you might resist the idea that biological systems is ergotic is for example uh the pretty plain fact that evolution isn't going anywhere right there doesn't seem to be some kind of evolutionary kind of attractor basin towards which uh all of evolution you know uh contra this kind of anthropocentric uh you know religious view of you know man's place in nature well humans aren't all that special um and we're uh we're just like one among uh you know a series of other uh animals um yeah so that wait one quick before we go to stephen also if your body temperature hits a certain very low or very high value it's not returning to a steady state either and so that's like there's a lot of reasons to think at a certain scale of analysis that there's violations of this kind of behavior stephen i mean very clearly we all die right yeah we all

dissipate and so at long enough time scales nothing is uh hasn't has like this non-equilibrium steady state or is ergodic in that sense right like we all we all just dissipate you know in some sense the the the only real ultimate ergotic you know end point is the heat death of the universe right so yeah it's uncanny valley because at a short time scale you're stable then at the lifelong time scale you dissipate and then again you're just a flash in the pan so very cool and highlights that it's important what scale we're talking about steven before we close out the slide yeah i also think that you know we talk about the system is the system ergodic internally externally but i don't know whether this is right but i kind of i'm more from a practice perspective trying to do the sense making of active inference and that kind of the dynamics of things going backwards and forwards so i i'm trying to think of the ergodicity as maybe more about the process of this noisy chaotic like muddy was saying data going backwards and forwards or across the markov blanket and within there there's ergodicity in that you know within and then that even if the systems i don't know whether they are or not ergodic of the organism but there's something in the data stream that allows you to extract information from the entropy the sort of the noise so i was trying to work out if you know that that might be two questions here i think it ends up going to this bigger question about whether the systems are ergodic but i'm not sure whether it's about the dynamics the non-linear dynamics have ergodicity sort of tucked away in there yeah just to give an ecological example let's think about a predator prey relationship like a lotka volterra model you know the cougars and the rabbits or something like that so it's always changing through time and there's a winnerless competition where they're always oscillating and so from the first pass the first moment it's not ergotic it's changing at certain time scales but it has what muddy was referring to as this dynamical stationarity or the dynamical ergodicity it's kind of like the more things change the more they stay the same sometimes when we're talking about these higher order relational dynamics or rates of change the partial derivatives that muddy mentioned then it's actually through time at the correct time scale of analysis that you see see that the predator prey relationship is in like this higher order ergodicity and then at another time scale above that the predator and the prey species don't even exist so of course it can't be an ergotic system so again we see that there's time scales where systems display ergodicity and time skills and spatial scales where they don't as well as relying on questions about exchange ability and other attributes but just in the last couple minutes this is really awesome i want to just run through the figures give people a little peek into some of the formalisms and the ideas that we'll be talking about further in 7.2 and this is a quotation about how exactly niche construction enters

the picture and especially in terms of epistemic resources which relates to the technological niche of humans as well as the skilled intentionality here's a good quotation about how in the multi-scale framework active inference is a group behavior that's a group activity and so this sort of connects the dots between active inference and collective because even when we draw the organismal loop with active inference we're talking about collectives of neurons or of different cell types and also ties it back to closed causality circular causality and synergetics this idea of coordinating slower order parameters and then rapidly oscillating higher order parameters and that's kind of like the dynamics of the gas cloud looks stationary visibly but if you went down there it'd be the seething mess and you'd say there's no way this is a stationary state but it is at a little bit of a higher level and then um this last quote was just how um at work one's behavior acts to justify their sensory states and their prior beliefs about them and then that kind of reminded me of a figure from the paper written with alex and sasha and others where we talked about that a little bit more specifically from a technological level um and just to quickly walk through the figures before we return figure one marco blanket it's gotta be figure one it's not it's not a ramstead paper if it isn't figure two is also unpacked a little bit in the answering schrodinger's question paper and this one there's what you see and there's what you don't see what you see is the potential explanatory scope across spatial and temporal scales that range over many orders of magnitude but what you don't see is that there's nothing that's at the for example one millisecond and one kilometer scale maybe a lightning strike is able to dissipate energy rapidly at that scale but it doesn't coordinate effectively it doesn't have that closed causality i should say uh but like i no longer agree with this figure uh casper has been i starting may 2018 uh i guess uh so casper uh was quick to point out that you don't you don't need a separate set of time scales the the spatial scales that we pointed out are really ontological scales so like the subcellular scale is a spatial scale and a temporal scale uh simultaneously so uh you know the cellular processes occur over a few milliseconds and yeah they have their specific size and these these scales like uh the the concomitant like associated spatial and temporal scales really increase along this diagonal uh kind of together um so i mean another thing that kind of breaks this uh apart is where is culture here right so culture is very fast uh but it's also very large scale a process so uh this is not a figure that i would recommend like learning by heart or anything i think it turns out to be wrong now but science progresses right so yep we were wrong i still find it interesting especially with that caveat that it's not like there's small things happening at the tiny scale and then there's no um relationship with the niche construction like niche construction for the cultural niche and the

communication niche is related to cellular processes so there's just what's happening but then it's interesting to think about how the functional levels of organization do align along this very vague qualitative diagonal but not a formal approach for sure here figure 3 is another recap figure having to do with i think something about broccoli and how it's a fractal now it's about markov blankets and about nested systems and how markov blankets at one level can be thought of as engaging in collective behavior at a higher level figure four again an example that we've seen before with looking at how morphology can arise as a function of initially equivalent cells acting morphologically to reduce their surprise about their expectations about the extracellular gradients and where they are and what that means for what the organism is doing and then how development and the evolution of development could reflect beliefs and action generative sequences allow the organism to fulfill functional morphologies figure five i don't think is one that we've seen yet um but it connects the active inference and a ford inside with a free energy minimization side and there's some mathematics there that i hope to be actually going into in the coming weeks as we've chosen the paper for eight which is going to be scaling active inference and we'll be looking to have a lot of discussion on the mathematics because um it's definitely one of the areas we want to dive deeper into and then the last figure is a partition of vector states into particles which are the the um the sets of balls here and um the blanket states so i'll go into that a lot next week this is probably the key figure perfect for the whole thing thanks for noting that we'll definitely go into this um in the uh last minute thanks for participating in this short and exciting 7.1 next time in 7.2 we're going to have more time to talk about some of the specific questions that have been arising as well as figure 6 as maxwell pointed out everyone live or in replay is welcome to send us comments or questions that would be cool to unpack live today i really appreciate just hearing about how these different ecosystem conceptualizations and ways of thinking about the same ecosystem same way of thinking across different ecosystems how do they all combine and we provide follow-up forms to all live participants so it's in the calendar invite and it'd be great if you could fill it out we also solicit feedback suggestions and questions from our audience and from our other participants who aren't live here and uh all we can say to that is thanks for the participation thanks everyone really an awesome discussion and just as always it's another level of understanding another coat of paint on the mural to hear from everyone's perspective so i deeply appreciate that and great times anyone want to give a uh singular closing thought to cap out the stream long live ergodicity muddy go for it oh i was just gonna say no exactly can i ask

you a question that's my observation perfect thanks everyone thanks again for listening for participating and we will see you next time